

CSE 4511: Computer Networks**QUIZ 2****DURATION: 25 MINUTES****WINTER SEMESTER, 2024-2025****FULL MARKS: 20****STUDENT ID:**

1. a) Mention the objectives of each DHCP messages.

4
(CO1)
(PO1)

Solution:

DHCP Discover locates available servers; Offer proposes IP/parameters; Request requests the offered IP or confirms a renewal; ACK confirms the lease allocation.

Rubric:

1. For each correct objective, 1 mark

- b) Which DHCP messages are optional and why?

2 + 2
(CO1)
(PO1)

Solution:

Discover and Offer are optional during lease renewal. If a client remembers its previous address, it can skip directly to sending a DHCP Request to extend the lease.

Rubric:

1. 1 mark for mentioning correct message
2. 1 mark for each reasoning

2. a) With a diagram briefly explain count to infinity problem

4
(CO1)
(PO1)

Solution:

It is a routing loop where routers continuously increment the hop count to a failed destination. It occurs because routers rely on neighbor rumors (Distance Vector) without knowing the full topology. **[Diagram not given]**

Rubric:

1. 2 marks for explanation
2. 2 marks for correct topology

- b) Does the problem happen in OSPF? Provide proper reasoning.

4
(CO1)
(PO1)

Solution:

No. OSPF is a Link State protocol where every router has a complete map (LSDB) of the network topology, allowing Dijkstra's algorithm to calculate loop-free paths locally.

Rubric:

1. 1 mark for stating no
2. 3 marks for reasoning

3. Differentiate between learning based forwarding and calculation based forwarding.

4
(CO1)
(PO2)

Solution:

Learning-based (switches) builds tables by passively observing source MACs of incoming frames. Calculation-based (routers) runs algorithms (like Dijkstra) to actively compute optimal paths. **[Other differentiations are not added here]**

Rubric:

1. 1 mark for each correct differences